

A Noncircular Oval with Convex Unit-Tangent Iterates

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Abstract

For an oriented regular plane curve γ , let $\mathcal{T}\gamma$ be obtained by moving each point one unit in the positive tangent direction. We prove that there is a smooth noncircular oval Γ_0 for which every iterate $\mathcal{T}^n\Gamma_0$ is an oval. This answers a question of Tabachnikov [6, Conjecture 4] in the negative. The construction begins with a complete convex hairpin C satisfying $\mathcal{T}C = C + (V, 0)$. Periodizing the sine of its steering angle gives exact centrally symmetric rear–front pairs $R_H \mapsto F_H$. After their half-perimeters are matched, a direct L^1 estimate compares their intrinsic curvature measures. A regularizing backward-shadowing theorem converts the resulting inverse pseudo-orbit into an exact infinite orbit.

1 Introduction

The unit-tangent transform moves each point of an oriented plane curve one unit in the positive tangent direction. Circles remain circles under this operation. We ask whether any noncircular oval has the same property.

Let

$$\gamma : \mathbb{R}/L\mathbb{Z} \longrightarrow \mathbb{R}^2$$

be a smooth regular closed curve, parametrized by arclength s , and let $\tau_\gamma = \gamma_s$ be its unit tangent. Define

$$(\mathcal{T}\gamma)(s) = \gamma(s) + \tau_\gamma(s). \tag{1}$$

The definition is independent of the regular parametrization. If the image is regular, we retain the point correspondence and orientation, then reparametrize by arclength before applying $\mathcal{T}$ again.

We orient convex curves counterclockwise, rotate the tangent counterclockwise through $\pi/2$ to obtain the inward normal ν , and write $\tau_s = k\nu$. An *oval* is a smooth embedded closed curve with $k > 0$ everywhere.

Theorem 1.1. *There is a noncircular oval Γ_0 such that $\mathcal{T}^n\Gamma_0$ is an oval for every integer $n \geq 0$.*

Tabachnikov conjectured that a curve with this property must be a circle [6, Conjecture 4]; Theorem 1.1 answers that question negatively. A circle of radius r is sent to one of radius $\sqrt{r^2 + 1}$. Our example instead becomes increasingly elongated. Its local model is a complete convex hairpin with parallel ends that $\mathcal{T}$ carries to a translate of itself.

Suppose $F = \mathcal{T}R$. We call R the *rear track*, F the *front track*, and the angle between corresponding tangents the *steering angle*. This is the classical bicycle correspondence; see [1, 2, 3, 5] for its geometry and history. A prescribed front does not determine a rear until the initial direction of the unit segment has been chosen. Once that direction is fixed, the first-order steering equation determines the rear track locally. For a closed front, closed rear tracks are periodic solutions, or

fixed points of the bicycle monodromy; there may be none, one, or several. In the low-curvature class used here, there is a unique periodic solution on the branch $0 < \delta < \pi/2$. We denote this selected inverse by $\mathcal{B}$; the notation refers only to this branch.

The proof has four steps.

1. An order-preserving operator and explicit barriers produce a complete strictly convex hairpin C with $\mathcal{T}C = C + (V, 0)$.
2. The steering mass $y = \sin \delta$ of this hairpin is a positive, exponentially localized pulse of mass π . Periodizing y gives exact centrally symmetric pairs $\mathcal{T}R_H = F_H$.
3. After matching half-perimeters, the rear curvature measure of R_H and the front curvature of $F_{P(H)}$ are both compared with the same periodization of the isolated hairpin curvature. Their L^1 difference is exponentially small.
4. Curvature interpolation turns these discrepancies into summable normal paths. Exact Jacobi estimates for $\mathcal{B}$ keep the paths in a uniform low-curvature class and yield a convergent sequence of terminal pullbacks.

Formalization. The complete proof of Theorem 1.1 has been formalized and machine-checked in the Lean 4 proof assistant on top of the Mathlib library. The formal development constructs every analytic input—the translating hairpin, the exact two-cap pairs, the curvature-measure matching, and the backward-shadowing limit—and derives the theorem unconditionally, with no unproved assumptions and no axioms beyond Lean’s standard foundations.

2 One tangent step: rear and front tracks

Let $F = \mathcal{T}R$. Parametrize F by front arclength s , write x for rear arclength, and choose tangent angles Θ, Ψ so that

$$\tau_F = (\cos \Theta, \sin \Theta), \quad \tau_R = (\cos \Psi, \sin \Psi).$$

Put $\delta = \Theta - \Psi$. Resolving the front velocity in the rear Frenet frame gives

$$\frac{dx}{ds} = \cos \delta, \quad \frac{d\Psi}{ds} = \sin \delta. \quad (2)$$

If $K = \Theta_s$ and $k = \Psi_x$ are the front and rear curvatures, then

$$K = \delta_s + \sin \delta, \quad k = \tan \delta. \quad (3)$$

If the front is strictly convex and $\phi = \Theta$ is used as parameter, with radius of curvature $q = 1/K$, then

$$\delta_\phi = 1 - q(\phi) \sin \delta. \quad (4)$$

These are the standard bicycle equations; see, for example, Levi–Tabachnikov [4]. Equation (4) also makes the preceding qualification precise: an initial value $\delta(\phi_0)$ determines a rear track, while a closed rear track is a periodic solution.

Lemma 2.1 (Low-curvature inverse). *Let F be a smooth strictly convex closed curve with*

$$0 < K \leq \kappa < 1.$$

There is a unique closed rear track on the branch $0 < \delta < \pi/2$. It is smooth and strictly convex, and

$$k_R \leq \frac{\kappa}{\sqrt{1 - \kappa^2}}. \quad (5)$$

If F is centrally symmetric, so is the selected rear track.

Proof. Set $d_0 = \arcsin \kappa$. In (4), the vector field is positive at 0 and nonpositive at d_0 . Thus the time- 2π map sends $[0, d_0]$ into itself and has a fixed point. If δ_1 and δ_2 are two solutions in this interval, their difference satisfies

$$w' = -q(\phi) \cos \xi(\phi) w$$

for a value $\xi(\phi)$ between $\delta_1(\phi)$ and $\delta_2(\phi)$. One full turn strictly decreases $|w|$, so the periodic solution is unique. Its rear tangent angle is strictly increasing, and

$$k_R = \tan \delta \leq \tan d_0 = \frac{\kappa}{\sqrt{1 - \kappa^2}}.$$

Half-turn symmetry follows from uniqueness. □

Lemma 2.2 (Convex consecutive tracks). *Let γ be a smooth closed convex curve. If $\mathcal{T}\gamma$ is also convex, then γ is strictly convex.*

Proof. Let $k \geq 0$ be the curvature of γ and put

$$u = \frac{k}{\sqrt{1 + k^2}}.$$

A Frenet calculation gives, at corresponding points,

$$K = u_x + u,$$

where $K \geq 0$ is the curvature of $\mathcal{T}\gamma$. Therefore

$$(e^x u(x))' = e^x K(x) \geq 0.$$

If $u(x_0) = 0$ at some point, periodicity would force $u \equiv 0$, contrary to the turning number of a closed convex curve. Hence $k > 0$. □

3 A translating hairpin

A *hairpin* is a complete embedded strictly convex curve whose tangent turns through π and whose two ends are asymptotic to parallel lines. We construct one satisfying $\mathcal{T}C = C + (V, 0)$.

Parametrize the curve by tangent angle $0 < \theta < \pi$ and write

$$\tau(\theta) = (\cos \theta, \sin \theta), \quad C_\theta = \rho(\theta)\tau(\theta), \quad f(\theta) = \rho(\theta) \sin \theta.$$

If $C = (X, Z)$, then

$$X' = f \cot \theta, \quad Z' = f. \quad (6)$$

The tangent angle of $C + \tau$ is

$$g(\theta) = \theta + d(\theta), \quad d(\theta) = \arctan \frac{\sin \theta}{f(\theta)}.$$

Lemma 3.1 (Translator equation). *Suppose that f is smooth on $(0, \pi)$, bounded above, and bounded below by a constant greater than 1. If*

$$\int_{\theta}^{\theta+d(\theta)} f(t) dt = \sin \theta, \quad d(\theta) = \arctan \frac{\sin \theta}{f(\theta)}, \quad (7)$$

then $g = \theta + d$ is an increasing diffeomorphism of $(0, \pi)$ and

$$C(\theta) + \tau(\theta) = C(g(\theta)) + (V, 0) \quad (8)$$

for a constant V .

Proof. Differentiating (7) gives

$$f(g)g' = f + \cos \theta > 0.$$

The integral identity gives the vertical component of (8). For the horizontal component, put $V(\theta) = X(\theta) + \cos \theta - X(g(\theta))$. Using $\tan(g - \theta) = \sin \theta / f(\theta)$ and the displayed formula for g' gives $V' = 0$. $\square$

For a bounded measurable function $f : (0, \pi) \rightarrow \mathbb{R}$ satisfying

$$1 < m \leq f \leq M < \infty,$$

let

$$A_f(t) = \int_0^t f(u) du.$$

For $0 < \theta < \pi$, define

$$D_f(\theta) = A_f^{-1}(A_f(\theta) + \sin \theta) - \theta. \quad (9)$$

This is well defined because

$$A_f(\pi) - A_f(\theta) \geq m(\pi - \theta) > \sin \theta.$$

Set

$$(\mathcal{P}f)(\theta) = \sin \theta \cot D_f(\theta), \quad 0 < \theta < \pi. \quad (10)$$

The endpoints 0 and π are not part of the domain. Excluding them allows the monotone limit to have an endpoint layer even though every finite iterate extends continuously to $[0, \pi]$.

Lemma 3.2 (Monotone translator operator). *The map $\mathcal{P}$ sends bounded profiles with $1 < m \leq f \leq M$ to positive continuous functions on $(0, \pi)$. It is order preserving:*

$$f \leq h \implies \mathcal{P}f \leq \mathcal{P}h.$$

A bounded profile f satisfies $f = \mathcal{P}f$ on $(0, \pi)$ if and only if it satisfies the translator equation (7) there.

Proof. The primitive A_f is continuous and strictly increasing, so D_f , and hence $\mathcal{P}f$, is continuous on the open interval. Also

$$0 < D_f(\theta) \leq \frac{\sin \theta}{m} < 1 < \frac{\pi}{2},$$

so $\mathcal{P}f > 0$.

If $f \leq h$, then, for fixed θ , the integral of h reaches $\sin \theta$ no later than the integral of f . Hence $D_h \leq D_f$, and the monotonicity of cotangent gives $\mathcal{P}f \leq \mathcal{P}h$.

The fixed-point equation $f = \mathcal{P}f$ is equivalent to

$$\tan D_f(\theta) = \frac{\sin \theta}{f(\theta)}.$$

Since $0 < D_f < \pi/2$, this says precisely that $D_f = d$ in (7). The definition of D_f now gives the integral equation. Conversely, that equation gives $D_f = d$, and hence $f = \mathcal{P}f$. $\square$

Put

$$u_0(\theta) = \frac{2}{3}(1 - \cos \theta), \quad v_-(\theta) = -\cos \theta, \quad v_+(\theta) = \cos \theta + 2,$$

and, for $\varepsilon > 0$, define

$$f_\varepsilon^\pm(\theta) = \varepsilon^{-1} + u_0(\theta) + \varepsilon v_\pm(\theta). \quad (11)$$

The barriers satisfy

$$f_\varepsilon^+(\theta) - f_\varepsilon^-(\theta) = 2\varepsilon(1 + \cos \theta) \geq 0.$$

For sufficiently small positive ε , their continuous extensions to $[0, \pi]$ satisfy

$$\min_{0 \leq \theta \leq \pi} f_\varepsilon^-(\theta) = \varepsilon^{-1} - \varepsilon, \quad \min_{0 \leq \theta \leq \pi} f_\varepsilon^+(\theta) = \varepsilon^{-1} + 3\varepsilon.$$

Thus both barriers belong to the domain of $\mathcal{P}$, and $f_\varepsilon^- \leq f_\varepsilon^+$.

Lemma 3.3 (Explicit barriers). *For all sufficiently small $\varepsilon > 0$,*

$$f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^+ \leq f_\varepsilon^+ \quad (0 < \theta < \pi). \quad (12)$$

Proof. For a profile f , set

$$d_f = \arctan \frac{\sin \theta}{f(\theta)}, \quad \mathcal{R}(f) = \int_\theta^{\theta+d_f} f(t) dt - \sin \theta.$$

The integral $D \mapsto \int_\theta^{\theta+D} f$ is strictly increasing. Since D_f is the value at which it equals $\sin \theta$, while $f(\theta) = \sin \theta \cot d_f$, one has

$$\mathcal{R}(f) \geq 0 \iff D_f \leq d_f \iff \mathcal{P}f(\theta) \geq f(\theta),$$

and similarly with the inequalities reversed. Thus the sign of $\mathcal{R}(f)$ is the sign of $\mathcal{P}f - f$. For $f = \varepsilon^{-1} + u_0 + \varepsilon v$, divide by the common second-order zero and expand in ε :

$$\frac{2\mathcal{R}(f)}{d_f^2 \sin \theta} = \varepsilon \left(\frac{v'(\theta)}{\sin \theta} + \frac{2}{9} \cos \theta \right) + O(\varepsilon^2), \quad (13)$$

where the quotient is interpreted continuously at 0 and π , and the remainder is uniform in θ .

An exact factorization controls the apparent endpoint singularities. Each barrier has the form

$$f(\theta) = A + B \cos \theta.$$

Write $c = \cos \theta$, $s = \sin \theta$, $F = A + Bc$, and $d = \arctan(s/F)$. Direct integration gives

$$\mathcal{R}(f) = Ad + B\{s(\cos d - 1) + c \sin d\} - s. \quad (14)$$

Since $F = s \cot d$, this becomes

$$\frac{2\mathcal{R}(f)}{d^2 s} = \Phi_0(d^2) + B\Phi_1(d^2) + \frac{Bc}{F}\Phi_2(d^2), \quad (15)$$

where

$$\begin{aligned} \Phi_0(r) &= \frac{2(\sqrt{r}/\tan\sqrt{r} - 1)}{r}, & \Phi_1(r) &= \frac{2(\cos\sqrt{r} - 1)}{r}, \\ \Phi_2(r) &= \frac{2(\sin\sqrt{r} - \sqrt{r})}{r \tan\sqrt{r}}. \end{aligned}$$

All three functions extend analytically through $r = 0$, with

$$\Phi_0(0) = -\frac{2}{3}, \quad \Phi_1(0) = -1, \quad \Phi_2(0) = -\frac{1}{3}.$$

If $v(\theta) = \alpha + \beta c$, then

$$F = \varepsilon^{-1}\{1 + \varepsilon u_0(c) + \varepsilon^2 v(c)\}, \quad B = -\frac{2}{3} + \varepsilon\beta.$$

Also,

$$\left(\frac{s}{F}\right)^2 = \frac{\varepsilon^2(1 - c^2)}{\{1 + \varepsilon u_0(c) + \varepsilon^2 v(c)\}^2}, \quad \frac{Bc}{F} = \frac{\varepsilon Bc}{1 + \varepsilon u_0(c) + \varepsilon^2 v(c)}.$$

Thus the right side of (15) is smooth in (ε, c) on a neighborhood of $[0, \varepsilon_0] \times [-1, 1]$. Its value at $\varepsilon = 0$ is zero, and its ε -derivative there is

$$-\beta + \frac{2}{9}c = \frac{v'(\theta)}{\sin\theta} + \frac{2}{9}\cos\theta.$$

Taylor's theorem on this compact rectangle proves (13) with a uniform remainder.

For $v_- = -\cos\theta$, the coefficient in parentheses is

$$1 + \frac{2}{9}\cos\theta \geq \frac{7}{9}.$$

For $v_+ = \cos\theta + 2$, it is

$$-1 + \frac{2}{9}\cos\theta \leq -\frac{7}{9}.$$

Thus $\mathcal{R}(f_\varepsilon^-) \geq 0$ and $\mathcal{R}(f_\varepsilon^+) \leq 0$ for sufficiently small ε . The middle inequality follows from Lemma 3.2. $\square$

Theorem 3.4 (Translating hairpin). *For every sufficiently small $\varepsilon > 0$, equation (7) has a solution $f_\varepsilon \in C^\infty(0, \pi)$ satisfying*

$$f_\varepsilon^- \leq f_\varepsilon \leq f_\varepsilon^+.$$

The corresponding curve is a complete embedded strictly convex hairpin in a strip of finite width, and

$$\mathcal{TC} = C + (V, 0)$$

for some $V > 0$.

Proof. Start with $f_0 = f_\varepsilon^-$ on $(0, \pi)$, and define $f_{n+1} = \mathcal{P}f_n$. The barrier lemma and monotonicity give

$$f_\varepsilon^- \leq f_0 \leq f_1 \leq \cdots \leq f_\varepsilon^+.$$

Let $f = \lim_n f_n$ pointwise on $(0, \pi)$, and set

$$A_n(t) = \int_0^t f_n(u) du, \quad A(t) = \int_0^t f(u) du.$$

Monotone convergence gives

$$\sup_{0 \leq t \leq \pi} |A_n(t) - A(t)| \leq \int_0^\pi (f - f_n) \longrightarrow 0.$$

The primitives have a common positive lower bound for their slopes. Their inverses therefore converge uniformly on compact subintervals of their common ranges. For every $0 < \theta < \pi$,

$$D_{f_n}(\theta) \longrightarrow A^{-1}(A(\theta) + \sin \theta) - \theta = D_f(\theta).$$

Passing to the limit in $f_{n+1}(\theta) = \sin \theta \cot D_{f_n}(\theta)$ gives $f = \mathcal{P}f$ on the open interval. Lemma 3.2 therefore gives the translator equation.

The primitive A is continuous and strictly increasing; hence D_f is continuous. The identity $f = \sin \theta \cot D_f$ then makes f continuous on $(0, \pi)$. The relation

$$A(\theta + D_f(\theta)) - A(\theta) = \sin \theta$$

may now be differentiated. Successive differentiations give $D_f, f \in C^\infty(0, \pi)$.

No endpoint values are assigned; the construction uses only the profile on $(0, \pi)$ and the uniform barrier bounds on that open interval ($f_\varepsilon^- \leq f \leq f_\varepsilon^+$ on $(0, \pi)$ only; the smooth extension off a collar, identically 1, is not required to satisfy the lower barrier). The constants C_j, D_j below depend on the fixed ε (quantifier order: first choose $\varepsilon \leq 1/40$, so that the curvature $\sin \theta / f_\varepsilon \leq 1/(\varepsilon^{-1} - \varepsilon) \leq 1/20$; then obtain $C_j(\varepsilon), D_j(\varepsilon)$); no uniform control of $\varepsilon D_j(\varepsilon)$ over ε is asserted.

Define C by (6). Its radius of curvature is $f / \sin \theta > 0$. Since $Z' > 0$, the curve is embedded. The barrier bounds imply that its arclength is infinite at both ends, that $X(\theta) \rightarrow -\infty$ at both ends, and that Z has finite one-sided limits. Hence the ends are asymptotic to parallel horizontal lines. Lemma 3.1 gives the translation law. At $\theta = \pi/2$, one has $g(\theta) > \theta$ and $\cot t < 0$ on the interval between them; therefore

$$V = - \int_{\pi/2}^{g(\pi/2)} f(t) \cot t dt > 0.$$

□

Fix such a hairpin so wide that its curvature is below $1/20$. Regard it as the rear track and its translate as the front track. Parametrize the front by arclength s , let $\delta(s)$ be the steering angle, and put

$$y(s) = \sin \delta(s), \quad c(s) = \cos \delta(s) = \sqrt{1 - y(s)^2}. \quad (16)$$

Choose front and rear arclength origins so that one positive function $K_* : \mathbb{R} \rightarrow (0, \infty)$ is the intrinsic curvature of both translated copies. If $x = x(s)$ is the corresponding rear arclength, then

$$x' = c, \quad K_*(s) = y + \frac{y'}{c}, \quad K_*(x(s)) = \frac{y}{c}. \quad (17)$$

Lemma 3.5 (Hairpin pulse estimates). *For the fixed wide hairpin above, the function y and all its derivatives decay exponentially at both ends. More precisely, there are $a_-, a_+ > 0$ such that, for every $j \geq 0$ (with $C_j = C_j(\varepsilon)$, $D_j = D_j(\varepsilon)$ depending on the fixed ε),*

$$|y^{(j)}(s)| \leq C_j e^{a_- s} \quad (s \leq 0), \quad |y^{(j)}(s)| \leq C_j e^{-a_+ s} \quad (s \geq 0),$$

and

$$|y^{(j)}(s)| \leq D_j y(s). \quad (18)$$

It also satisfies

$$\int_{\mathbb{R}} y(s) ds = \pi, \quad \Delta := \int_{\mathbb{R}} (1 - c(s)) ds > 0, \quad (19)$$

and

$$K_*(s) \geq b_0 y(s) \quad (20)$$

for some constant $b_0 > 0$.

Proof. Tail bounds. Let u be rear arclength, let $\theta(u)$ be the rear tangent angle, and write

$$\kappa(u) = \theta_u = \frac{\sin \theta(u)}{f(\theta(u))}.$$

The barrier enclosure gives fixed constants $1 < m < M$ with $m \leq f \leq M$. The half-angle variable

$$r(u) = \tan \frac{\theta(u)}{2}$$

satisfies the exact equation

$$(\log r)_u = \frac{\theta_u}{\sin \theta} = \frac{1}{f(\theta)}. \quad (21)$$

Hence r , $\sin \theta$, and κ have two-sided exponential bounds at the two ends. In particular, κ^2 is integrable; indeed,

$$\int_{\mathbb{R}} \kappa(u)^2 du = \int_0^\pi \frac{\sin \theta}{f(\theta)} d\theta < \infty.$$

The shifted curvature equation. Let $\sigma(u)$ be front arclength at the corresponding point. Since

$$\sigma'(u) = \sqrt{1 + \kappa(u)^2},$$

the function $\sigma(u) - u$ is bounded. Choose the front and rear arclength origins so that the intrinsic curvature of both translated copies is the same function K_* . Then $K_*(u) = \kappa(u)$, and the Frenet formula for the front curvature is

$$K_*(\sigma(u)) = \frac{K'_*(u) + K_*(u) + K_*(u)^3}{(1 + K_*(u)^2)^{3/2}}. \quad (22)$$

Relative derivatives. We prove that

$$|K_*^{(j)}(u)| \leq C_j K_*(u) \quad (j \geq 0). \quad (23)$$

Equation (21) first gives a quantitative bounded-shift Harnack estimate. For every $D > 0$, there is C_D such that

$$C_D^{-1} K_*(u) \leq K_*(u + h) \leq C_D K_*(u) \quad (|h| \leq D). \quad (24)$$

To prove this, note that $r(u+h)/r(u)$ lies in a compact subinterval of $(0, \infty)$, depending only on D, m, M . The function $2r/(1+r^2) = \sin \theta$ changes by a bounded factor under such a multiplicative change of r , and $m \leq f \leq M$. Since $\sigma(u) - u$ is bounded, (24) gives

$$K_*(\sigma(u)) \leq CK_*(u).$$

Solving (22) for K'_* now gives $|K'_*| \leq CK_*$.

We spell out the induction. Assume (23) through order j . From $\sigma' = \sqrt{1 + K_*^2}$, every derivative $\sigma^{(r)}$, $1 \leq r \leq j+1$, is uniformly bounded; for $r \geq 2$ it is a finite sum of products involving derivatives of K_* of order at most $r-1$. Faà di Bruno's formula gives

$$\partial_u^q(K_* \circ \sigma) = \sum_{\ell=1}^q K_*^{(\ell)}(\sigma(u)) B_{q,\ell}(\sigma', \dots, \sigma^{(q-\ell+1)}),$$

where the $B_{q,\ell}$ are Bell polynomials. By the induction hypothesis, (24), and the preceding bounds on $\sigma^{(r)}$, every term is $O(K_*(u))$ for $q \leq j$.

Differentiate

$$K'_* = (1 + K_*^2)^{3/2} K_* \circ \sigma - K_* - K_*^3$$

exactly j times. Derivatives of the prefactor $(1 + K_*^2)^{3/2}$ are bounded; every term of positive derivative order contains at least one derivative of K_* and is therefore $O(K_*)$. The composed factor is $O(K_*)$ by the previous paragraph, and every derivative of K_*^3 is also $O(K_*)$. Hence $|K_*^{(j+1)}| \leq C_{j+1} K_*$, which closes the induction. The exponential bounds for K_* therefore extend to every derivative.

The steering pulse. Let $s = \sigma(u)$, and let $x(s) = u$ be the inverse correspondence. Since $\tan \delta = K_*(u)$,

$$y(s) = \sin \delta(s) = \frac{K_*(x(s))}{\sqrt{1 + K_*(x(s))^2}}, \quad c(s) = \frac{1}{\sqrt{1 + K_*(x(s))^2}}. \quad (25)$$

The derivatives of $x = \sigma^{-1}$ are uniformly bounded and, beyond first order, are smooth expressions in derivatives of K_* . Combining (25) with (23) and the chain rule gives (18) and exponential decay of every derivative of y .

Mass and the lower comparison. The mass identity is exact:

$$y(s) ds = \frac{K_*(u)}{\sqrt{1 + K_*(u)^2}} \sqrt{1 + K_*(u)^2} du = K_*(u) du.$$

The rear tangent turns through π , so $\int_{\mathbb{R}} y = \pi$. The integral defining Δ is finite because $1 - c = O(y^2)$, and it is positive because $y \not\equiv 0$.

By (23), $|(\log K_*)'| \leq C$, and $|s - x(s)|$ is bounded. The exact identity

$$K_*(x(s)) = \frac{y(s)}{c(s)}$$

therefore gives

$$K_*(s) \geq e^{-C|s-x(s)|} K_*(x(s)) \geq b_0 y(s)$$

for a fixed $b_0 > 0$, which proves (20). $\square$

4 Exact two-cap pairs

For $H > 0$, periodize the steering mass:

$$Y_H(s) = \sum_{m \in \mathbb{Z}} y(s - mH). \quad (26)$$

For all sufficiently large H , $0 < Y_H < 1$. Set

$$\delta_H = \arcsin Y_H, \quad c_H = \sqrt{1 - Y_H^2}, \quad (27)$$

$$K_H = Y_H + \frac{Y_H'}{c_H}, \quad k_H(x_H(s)) = \frac{Y_H(s)}{c_H(s)}, \quad x_H' = c_H. \quad (28)$$

The next lemma controls periodization and tail overlap, including their derivatives with respect to the period.

Lemma 4.1 (Exponential periodization). *Let $z > 0$ be smooth, exponentially decaying with all derivatives, and suppose $|z^{(j)}| \leq C_j z$. Put*

$$Z_H(s) = \sum_{m \in \mathbb{Z}} z(s - mH), \quad I_H = [-H/2, H/2].$$

For every pair of nonnegative integers r, q , there are $C_{r,q}, \beta_{r,q} > 0$ such that

$$\|\partial_H^q (Z_H - z)\|_{C^r(I_H)} \leq C_{r,q} e^{-\beta_{r,q} H}. \quad (29)$$

Here s is held fixed when H is differentiated. Furthermore, for fixed j, k ,

$$\sum_{m \neq n} |z^{(j)}(s - mH)| |z^{(k)}(s - nH)| \leq C e^{-\beta H} Z_H(s). \quad (30)$$

This relative overlap estimate also remains valid after any fixed number of H -derivatives.

Proof. For $s \in I_H$ and $m \neq 0$,

$$|s - mH| \geq (|m| - 1/2)H.$$

Termwise differentiation gives

$$\partial_H^q \partial_s^r z(s - mH) = (-m)^q z^{(r+q)}(s - mH).$$

The exponential tails dominate the polynomial factor $|m|^q$, proving (29).

For the overlap estimate, write $n = m + \ell$. The two pulse centers are $|\ell|H$ apart. Hence at least one of $s - mH$ and $s - nH$ lies at least $|\ell|H/2$ into a tail. The relative derivative bounds reduce each fixed derivative to the estimate for $z_m z_n$. Summing first over m , then over $\ell \neq 0$, proves the claim. Differentiation in H introduces only polynomial factors in the translation indices, and the same geometric series absorbs them. $\square$

Lemma 4.2 (Front periodization error). *For a period L , put*

$$y_m(s) = y(s - mL), \quad Y_L = \sum_m y_m, \quad G(z) = \frac{1}{\sqrt{1 - z^2}}.$$

Then, for all sufficiently large L ,

$$\left| K_L(s) - \sum_m K_*(s - mL) \right| \leq C \sum_{m \neq n} y_m(s) y_n(s) \leq C e^{-\beta L} Y_L(s). \quad (31)$$

Proof. The isolated identity gives

$$K_*(s - mL) = y_m(s) + G(y_m(s))y'_m(s),$$

whereas

$$K_L = Y_L + G(Y_L)Y'_L.$$

Therefore

$$K_L - \sum_m K_*(s - mL) = \sum_m y'_m (G(Y_L) - G(y_m)).$$

For large L , the functions Y_L and y_m take values in a fixed interval $[0, a]$, where $a < 1$. The function G is uniformly Lipschitz on this interval, and $|y'_m| \leq Dy_m$. Hence

$$\begin{aligned} \left| K_L - \sum_m K_*(s - mL) \right| &\leq C \sum_m y_m \sum_{n \neq m} y_n \\ &= C \sum_{m \neq n} y_m y_n. \end{aligned}$$

The second inequality in (31) is (30). □

Proposition 4.3 (Exact two-cap pairs). *For all sufficiently large H , equations (26)–(28) define smooth strictly convex centrally symmetric closed curves R_H, F_H such that*

$$\mathcal{T}R_H = F_H.$$

The front perimeter is $2H$, the rear perimeter is $2P(H)$, where

$$P(H) = \int_0^H c_H(s) ds, \tag{32}$$

and

$$P(H) = H - \Delta + O(e^{-\beta H}), \quad P'(H) = 1 + O(e^{-\beta H}). \tag{33}$$

If the translating hairpin is chosen sufficiently wide, one constant $\kappa_0 < 1$ bounds the curvatures of both closed curves for all sufficiently large H .

Proof. Choose Θ_H with $\Theta'_H = K_H$, and define the unit-speed front by $F'_H = \tau(\Theta_H)$. Because δ_H is periodic and $\int_0^H Y_H = \pi$,

$$\int_0^H K_H ds = \int_0^H (\delta'_H + Y_H) ds = \pi.$$

The front tangent reverses after one H -period. Hence the displacement during the second period is the negative of the first, and F_H closes after two periods. Set $\Psi_H = \Theta_H - \delta_H$ and

$$R_H = F_H - \tau(\Psi_H).$$

Then $\Psi'_H = Y_H$ and

$$R'_H = c_H \tau(\Psi_H),$$

so R_H is the rear track with curvature k_H , has half-perimeter $P(H)$, and also closes after two periods. Both curves are centrally symmetric.

The rear curvature is positive. Lemma 4.2 gives

$$K_H(s) = \sum_m K_*(s - mH) + E_H(s), \quad |E_H(s)| \leq Ce^{-\beta H} Y_H(s).$$

Since $K_* \geq b_0 y$,

$$K_H(s) \geq (b_0 - Ce^{-\beta H}) Y_H(s) > 0$$

for all sufficiently large H . A closed locally convex curve of turning number one is the boundary of a strictly convex body; hence both tracks are embedded ovals.

Set

$$\Phi(z) = 1 - \sqrt{1 - z^2}.$$

On the centered cell,

$$H - P(H) = \int_{-H/2}^{H/2} \Phi(Y_H(s)) ds.$$

Equation (29), smoothness of Φ on a fixed interval $[0, a]$ with $a < 1$, and the omitted exponential tails give

$$H - P(H) = \int_{\mathbb{R}} \Phi(y(s)) ds + O(e^{-\beta H}) = \Delta + O(e^{-\beta H}).$$

To differentiate, hold s fixed in the centered integral:

$$\begin{aligned} \frac{d}{dH}(H - P(H)) &= \frac{1}{2} \Phi(Y_H(H/2)) + \frac{1}{2} \Phi(Y_H(-H/2)) \\ &\quad + \int_{-H/2}^{H/2} \Phi'(Y_H(s)) \partial_H Y_H(s) ds. \end{aligned}$$

The endpoint terms are exponentially small. The last integral is also $O(e^{-\beta H})$ by (29), after replacing β by a smaller positive constant. This yields the derivative estimate in (33).

Choose the translating hairpin so wide that its intrinsic curvature is bounded well below one. The periodization and overlap estimates then keep both K_H and k_H below one fixed $\kappa_0 < 1$ for all sufficiently large H . $\square$

Lemma 4.4 (Uniform transverse width). *After a rigid motion, let the tangent angle of F_H on the centered half-period increase from 0 to π . Its displacement perpendicular to the asymptotic sides is*

$$W_H = \int_{-H/2}^{H/2} \sin \Theta_H(s) ds.$$

There is a constant C such that

$$0 < W_H \leq C$$

for all sufficiently large H .

Proof. Let Θ_* be the tangent angle of the isolated translated hairpin, normalized by $\Theta_*(-\infty) = 0$. The periodization estimates and the formula for K_H give, on the centered cell,

$$\|K_H - K_*\|_{\infty} \leq Ce^{-\beta H}.$$

Also $\Theta_*(-H/2) = O(e^{-\beta H})$. With $\Theta_H(-H/2) = 0$, integration therefore gives

$$\|\Theta_H - \Theta_*\|_{L^{\infty}(I_H)} \leq CH e^{-\beta H}.$$

Consequently, for any $0 < \beta' < \beta$ and all large H , $H\|\Theta_H - \Theta_*\|_\infty \leq CH^2e^{-\beta H} \leq C'e^{-\beta'H} \leq 1$, so

$$\begin{aligned} W_H &\leq \int_{I_H} \sin \Theta_*(s) ds + H\|\Theta_H - \Theta_*\|_\infty \\ &\leq \int_{\mathbb{R}} \sin \Theta_*(s) ds + 1 \end{aligned}$$

for all large H . The last integral is the finite strip width of the translating hairpin. $\square$

5 Curvature-measure matching

The isolated rear and front are translates, so their intrinsic curvatures are the same function K_* after independent choices of arclength origin. Use the front origin to fix the phase of y and its periodizations, and set $x_H(0) = x(0)$ for the closed rear. For a period L , write

$$\bar{K}_L(u) = \sum_{j \in \mathbb{Z}} K_*(u - jL).$$

Lemma 4.2 already compares the front curvature K_L with $\bar{K}_L$. We compare the rear curvature measure with the same profile.

Lemma 5.1 (Common phase and fundamental interval). *Let $I_H = [-H/2, H/2]$ and $J_H = x_H(I_H)$. Then J_H has length $P(H)$, its translates by $P(H)\mathbb{Z}$ tile $\mathbb{R}$, and both the rear and front comparisons use $\bar{K}_{P(H)}$ with the same central phase.*

Proof. Since $x'_H = c_H > 0$, the image J_H is an interval and

$$|J_H| = \int_{I_H} c_H(s) ds = P(H).$$

Its translates therefore tile the line. The choices of intrinsic origins and $x_H(0) = x(0)$ identify the central isolated term in both comparisons with K_* . $\square$

Set $P = P(H)$. The infinite-time argument uses only the next estimate.

Theorem 5.2 (Curvature-measure matching). *There are constants $C, \beta > 0$ such that*

$$\int_{\mathbb{R}/P(H)\mathbb{Z}} |k_H(u) - K_{P(H)}(u)| du \leq Ce^{-\beta H} \quad (34)$$

for all sufficiently large H .

Proof. Put $\alpha = \min(a_-, a_+)$. Exponential tails imply, for every $0 < \beta < \alpha$,

$$\int_{\mathbb{R}} y(s)y(s-r) ds \leq C_\beta e^{-\beta|r|}. \quad (35)$$

The integrand is bounded by $Ce^{-\alpha(|s|+|s-r|)}$. Consequently, for pulses $y_m(s) = y(s - mL)$,

$$\int_{-L/2}^{L/2} \sum_{m \neq n} y_m(s)y_n(s) ds \leq Ce^{-\beta L}. \quad (36)$$

This follows by writing $n = m + q$, unfolding the sum over m to $\mathbb{R}$, and summing (35) over $q \neq 0$.

We compare both curvatures with $\bar{K}_P$.

Let $I_H = [-H/2, H/2]$, use the phase convention above, and put $J_H = x_H(I_H)$. The interval J_H has length P . Positivity and the tail bounds give

$$\|Y_H - y\|_{L^1(I_H)} + \|Y_H - y\|_{L^\infty(I_H)} \leq Ce^{-\alpha H/2}.$$

Since $c_H = \sqrt{1 - Y_H^2}$ and $c = \sqrt{1 - y^2}$ stay uniformly positive,

$$\|c_H - c\|_{L^1(I_H)} \leq Ce^{-\alpha H/2}, \quad \|x_H - x\|_{L^\infty(I_H)} \leq Ce^{-\alpha H/2}.$$

The identity $K_*(x(s)) = y(s)/c(s)$ and the exact measure formula

$$k_H(x_H(s)) dx_H = Y_H(s) ds \tag{37}$$

now give

$$\begin{aligned} & \|k_H(u) du - \bar{K}_P(u) du\|_{\text{TV}} \\ &= \int_{I_H} |Y_H - c_H \bar{K}_P(x_H)| ds. \end{aligned}$$

Split the integrand as

$$\begin{aligned} Y_H - c_H \bar{K}_P(x_H) &= (Y_H - y) + (y - c_H K_*(x_H)) \\ &\quad - c_H \sum_{j \neq 0} K_*(x_H - jP). \end{aligned}$$

The first term is exponentially small. For the second, use $y = cK_*(x)$, boundedness of K_* and K'_* , and the preceding estimates. For some $\beta > 0$,

$$\int_{I_H} |y - c_H K_*(x_H)| ds \leq CH e^{-\alpha H/2} \leq Ce^{-\beta H}.$$

For the last term, change variables $u = x_H(s)$:

$$\begin{aligned} \int_{I_H} c_H \sum_{j \neq 0} K_*(x_H - jP) ds &= \int_{J_H} \sum_{j \neq 0} K_*(u - jP) du \\ &= \int_{\mathbb{R} \setminus J_H} K_*(v) dv. \end{aligned}$$

Lemma 5.1 gives the last equality. We have $x_H(s) = s + O(1)$ and $P = H + O(1)$. Hence the endpoints of J_H are $\pm H/2 + O(1)$, and the exponential tails of K_* make the last integral exponentially small. Therefore

$$\|k_H - \bar{K}_P\|_{L^1(\mathbb{R}/P\mathbb{Z})} \leq Ce^{-\beta H}. \tag{38}$$

For the front of period $L = P$, Lemma 4.2 and (36) yield

$$\|K_P - \bar{K}_P\|_{L^1(\mathbb{R}/P\mathbb{Z})} \leq Ce^{-\beta P} \leq C'e^{-\beta' H}. \tag{39}$$

Combining (38) and (39) proves (34). $\square$

6 Regularizing backward shadowing

Throughout this section, curves are centrally symmetric, centered at the origin, and supplied with a marked point and tangent direction. These markings fix intrinsic-arclength phase and rotation.

Lemma 6.1 (Compatible markings). *Let $\mathcal{T}R = F$ be a centered centrally symmetric rear–front pair, and fix an intrinsic point of R . Given a centered marked curve G , there is a unique rotation of the whole pair for which the tangent of R at the chosen point agrees with the marked tangent of G . This rotation preserves $\mathcal{T}R = F$, all intrinsic curvature functions, and every estimate below. A constant shift of arclength changes only the marked phase and leaves W and the quantities S_j unchanged.*

Proof. Rotate R and F by the same angle. The unit-tangent transform commutes with rigid motions, whereas intrinsic curvature and arclength are rotation invariant. A constant arclength shift is a reparametrization, so it changes neither the geometric path nor any normal-variation norm. $\square$

We use Lemma 6.1 whenever consecutive model curves are compared: rotate the whole later pair so that the marked tangent of its selected rear agrees with the marked tangent of the earlier front.

For a smooth path, write

$$X_t = \xi\tau + \eta\nu.$$

The tangential component only changes the parametrization; the normal component η changes the geometric curve. Set

$$W(\Gamma) = \int_0^1 \|\eta_t\|_{L^1} dt, \quad S_j(\Gamma) = \int_0^1 \|\partial_s^j \eta_t\|_{L^\infty} dt.$$

In this section, *marked geometric C^2 convergence* means uniform convergence, in one transported periodic parameter, of

$$X, \quad \tau, \quad \kappa, \quad g = |X_u|,$$

with the limiting speed g bounded away from zero. Paths may be piecewise smooth in the path parameter. The functionals W, S_0, S_1, S_2 are additive under concatenation, and rounding finitely many path corners changes them by an arbitrarily small amount.

Parametrize a countable concatenation on subintervals whose endpoints accumulate only at $t = 1$. Summability of the S_j -lengths gives a continuous extension at $t = 1$; equivalently, the concatenation is an absolutely continuous path in the marked geometric topology.

Lemma 6.2 (Curvature interpolation). *Let G_0, G_1 be marked centrally symmetric ovals with the same half-perimeter L and the same marked tangent direction. Suppose their intrinsic half-periodic curvatures $\kappa^{(0)}, \kappa^{(1)}$ satisfy $0 \leq \kappa^{(i)} \leq \kappa_* < 1$ for $i = 0, 1$ (nonnegativity suffices; strict positivity is not required). There is a constant-speed ($g \equiv 1$) path Γ from G_0 to G_1 whose curvature is at most κ_* and such that*

$$W(\Gamma) + S_0(\Gamma) + S_1(\Gamma) \leq C(1 + L)^2 \|\kappa^{(1)} - \kappa^{(0)}\|_{L^1(0, L)}. \quad (40)$$

The constant C depends only on κ_* and not on L .

Proof. Set $\kappa_t = (1 - t)\kappa^{(0)} + t\kappa^{(1)}$. Let $\theta_t(0)$ be the common marked tangent angle, and define

$$\theta_t(s) = \theta_t(0) + \int_0^s \kappa_t(r) dr.$$

For $0 \leq s \leq L$, put

$$X_t(s) = \int_0^s \tau(\theta_t(r)) dr - \frac{1}{2} \int_0^L \tau(\theta_t(r)) dr, \quad (41)$$

and extend by

$$X_t(s+L) = -X_t(s).$$

Because $\int_0^L \kappa_t = \pi$, this extension is smooth. Formula (41) makes closure, central symmetry, centering, and the marked tangent phase explicit. Its curvature is κ_t , so the whole path is strictly convex and has curvature at most κ_* .

Set $\epsilon = \|\kappa^{(1)} - \kappa^{(0)}\|_{L^1}$. Then $\|\dot{\theta}_t\|_\infty \leq \epsilon$, and differentiating (41) gives $\|\dot{X}_t\|_\infty \leq (3/2)L\epsilon$. If $\dot{X}_t = \xi\tau + \eta\nu$, unit-speed parametrization gives $\theta_t = \eta_s + \kappa_t\xi$. Hence

$$\|\eta\|_\infty \leq CL\epsilon, \quad \|\eta_s\|_\infty \leq C(1+L)\epsilon.$$

Integrating over the full perimeter proves (40). $\square$

In the shadowing scheme of Theorem 6.8, the factor $(1+L)^2$ in (40) is carried by the defect quantities e_n ; the tube constants $M_0, M_1, C_{\text{tube}}$ introduced there depend only on the curvature ceiling.

Lemma 6.3 (Smooth dependence of the selected rear). *Let F_t , $0 \leq t \leq 1$, be a smooth path of strictly convex closed fronts whose curvatures satisfy $0 < K_t \leq \hat{\kappa} < 1$. Then their selected rear tracks form a smooth path.*

Proof. Parametrize each front by its tangent angle $\phi \in \mathbb{R}/2\pi\mathbb{Z}$, and let $q_t = 1/K_t$ be its radius of curvature. The selected steering angle satisfies

$$\delta_\phi = 1 - q_t(\phi) \sin \delta.$$

Its periodic linearization is

$$w_\phi + q_t \cos \delta w = f.$$

On the selected strip $0 \leq \delta \leq \arcsin \hat{\kappa}$, the zeroth-order coefficient is bounded below by

$$\hat{\kappa}^{-1} \sqrt{1 - \hat{\kappa}^2} > 0.$$

The periodic linear operator is invertible, with a positive Green kernel and uniform bounds. The hyperbolic fixed point of the Poincaré map therefore depends smoothly on t , as does the reconstructed rear track. $\square$

Lemma 6.4 (Inverse Jacobi estimates). *Fix $\hat{\kappa} < 1$. Let Γ be a smooth path of strictly convex fronts of curvature at most $\hat{\kappa}$, and let $\mathcal{B}\Gamma$ be the path of selected rears. There are constants depending only on $\hat{\kappa}$ such that*

$$W(\mathcal{B}\Gamma) \leq W(\Gamma), \quad (42)$$

$$S_0(\mathcal{B}\Gamma) \leq C_0 W(\Gamma), \quad (43)$$

$$S_1(\mathcal{B}\Gamma) \leq C_1 (W(\Gamma) + S_0(\Gamma)), \quad (44)$$

$$S_2(\mathcal{B}\Gamma) \leq C_2 (W(\Gamma) + S_0(\Gamma) + S_1(\Gamma)). \quad (45)$$

The selected inverse gains one derivative: a C^r front, $r \geq 2$, has a C^{r+1} selected rear.

Proof. Let R_t be the rear path, parametrize $R = R_0$ by rear arclength x , and write $\dot{R} = \xi\tau + \eta_R\nu$. Put $F = R + \tau$ and let η_F be the front normal velocity. Since $\dot{\tau} = (\eta_{R,x} + k\xi)\nu$ and $\nu_F = -\sin\delta\tau + \cos\delta\nu$, the tangential terms cancel:

$$(1 + \partial_x)\eta_R = \sec\delta\eta_F \circ s. \quad (46)$$

On a rear circle of length ℓ , the periodic inverse of $1 + \partial_x$ is

$$(\mathcal{R}_\ell f)(x) = \frac{1}{1 - e^{-\ell}} \int_{x-\ell}^x e^{-(x-t)} f(t) dt.$$

Its kernel is positive and has integral one. Because $dx = \cos\delta ds$, (42) follows.

Lemma 2.1 bounds δ away from $\pi/2$ and gives a uniform upper bound for the rear curvature. Total turning then bounds the rear perimeter away from zero. Hence $\mathcal{R}_\ell$ maps L^1 uniformly into L^∞ , which proves (43). From

$$\eta_{R,x} = \sec\delta\eta_F \circ s - \eta_R$$

we get (44). Differentiating once gives

$$\eta_{R,xx} = (\sec\delta)_x \eta_F \circ s + \sec^2\delta (\eta_F)_s \circ s - \eta_{R,x}.$$

The identity $\delta_x = \sec\delta(K - \sin\delta)$ bounds the coefficient uniformly and proves (45).

For the regularity assertion, the rear tangent angle in front arclength satisfies $\Psi_s = \sin(\Theta - \Psi)$. If the front is C^r , then Θ is C^{r-1} , the equation gives $\Psi \in C^r$, and one integration in rear arclength gives a C^{r+1} rear curve. $\square$

Lemma 6.5 (Completeness of summable normal paths). *Let G_j be marked regular curves joined by piecewise smooth paths Γ_j . Suppose all intermediate curvatures are bounded by a common constant and*

$$\sum_j (S_0(\Gamma_j) + S_1(\Gamma_j) + S_2(\Gamma_j)) < \infty.$$

Then G_j converges in marked geometric C^2 to a regular C^2 curve.

Proof. Concatenate the paths, rephasing the next path at each junction so that its parameter agrees with the one transported from the preceding path; the quantities S_j are unchanged by such a constant arclength shift. If the original velocity is $\xi\tau + \eta\nu$ and the speed in a periodic parameter is g , solve

$$\phi_t(t, u) = -\frac{\xi(t, \phi(t, u))}{g(t, \phi(t, u))}, \quad \phi(0, u) = u.$$

The reparametrized path has velocity $\eta\nu$. This transports the chosen parameter and changes neither the geometric path nor its normal component. We may therefore assume

$$X_t = \eta\nu.$$

If u is the transported periodic parameter and $g = |X_u|$, then, for our inward normal convention,

$$\tau_t = \eta_s\nu, \quad \kappa_t = \eta_{ss} + \kappa^2\eta, \quad (\log g)_t = -\kappa\eta. \quad (47)$$

These formulas use the inward-normal convention fixed in the introduction.

The assumed summability makes X , τ , κ , and $\log g$ uniformly Cauchy along the concatenated path. Since the total variation of $\log g$ is finite and every initial speed is positive, the limiting speed g is positive. Passing to the limit in

$$X_u = g\tau, \quad \tau_u = g\kappa\nu$$

shows that the limit is a regular C^2 curve. The transported marker and tangent direction converge as part of the same parametrized convergence. $\square$

Lemma 6.6 (Stopped curvature estimate). *Let $\Gamma = (C_t)_{0 \leq t \leq 1}$ be a smooth path written in normal gauge, $X_t = \eta\nu$. Suppose*

$$\max \kappa_{C_0} \leq \kappa_b < \kappa_e$$

and

$$S_2(\Gamma) + \kappa_e^2 S_0(\Gamma) < \kappa_e - \kappa_b.$$

Then $\max \kappa_{C_t} < \kappa_e$ for every t .

Proof. If T were the first exit time, then on $[0, T]$ the inward-normal flow identity would give

$$\|\kappa(T) - \kappa(0)\|_\infty \leq \int_0^T (\|\eta_{ss}\|_\infty + \kappa_e^2 \|\eta\|_\infty) dt < \kappa_e - \kappa_b,$$

contradicting $\max \kappa(T) = \kappa_e$. $\square$

Lemma 6.7 (Selected inverse on the closed strip). *Let F be a regular convex C^2 closed curve with $0 \leq K \leq \widehat{\kappa} < 1$. There is a unique periodic steering solution*

$$0 \leq \delta \leq \arcsin \widehat{\kappa}$$

of $\delta_s = K - \sin \delta$. It defines a regular convex rear track. After normalized-arclength identification of the parameter circles, this rear depends continuously in C^2 on the front. If the front is C^r , $r \geq 2$, the rear is C^{r+1} . Moreover the steering angle is 2-Lipschitz in front arclength ($|\delta'| = |K - \sin \delta| \leq 2$), and the inverse of the rear-arclength correspondence $x = x(s)$, the front arclength $\sigma = x^{-1}$ as a function of rear arclength, satisfies

$$|\sigma(u) - \sigma(v)| \leq \frac{|u - v|}{\sqrt{1 - \widehat{\kappa}^2}}.$$

Proof. The vector field points into the displayed interval, so its Poincaré map has a fixed point. The difference of two solutions satisfies $w_s + c(s)w = 0$, where $c(s) \geq \sqrt{1 - \widehat{\kappa}^2} > 0$; periodicity forces $w = 0$. Also $x_s = \cos \delta \geq \sqrt{1 - \widehat{\kappa}^2}$, so the rear is regular and has nonnegative curvature $\tan \delta$; inverting this bound gives the Lipschitz estimate for $\sigma = x^{-1}$, and the 2-Lipschitz bound for δ is immediate from the steering equation. When the perimeter varies, write the equation in normalized arclength. Continuous dependence for this scalar periodic ODE then gives convergence of δ , the coordinate change x , and the reconstructed rear. In front arclength $\Psi_s = \sin(\Theta - \Psi)$; the same derivative count as in Lemma 6.4 gives the asserted regularity gain. $\square$

Theorem 6.8 (Regularizing backward shadowing). *Let Q_n be marked centrally symmetric ovals, and put $A_n = \mathcal{B}Q_{n+1}$. Assume that Q_n and A_n have the same half-perimeter L_n and the same marked tangent direction. Assume also that all their curvatures are bounded by one constant $\kappa_0 < 1$. Define*

$$e_n = (1 + L_n)^2 \|\kappa_{Q_n} - \kappa_{A_n}\|_{L^1(0, L_n)}, \quad r_n = \sum_{m=n}^{\infty} e_m.$$

Suppose that

$$r_0 < \infty. \tag{48}$$

After discarding finitely many initial indices, the terminal pullbacks

$$Z_n^{(N)} = \mathcal{B}^{N-n} Q_N$$

are defined for all $N \geq n$ and converge in geometric C^2 to curves X_n satisfying

$$X_n = \mathcal{B}X_{n+1}, \quad \mathcal{T}X_n = X_{n+1}.$$

Every X_n is a smooth oval. On the retained tail, a constant $C = C(\kappa_0)$, independent of n and N , satisfies

$$d_H(X_n, Q_n) + |\text{Per}(X_n) - 2L_n| \leq Cr_n. \quad (49)$$

Here d_H denotes Hausdorff distance. For every $N \geq n$, there is a concatenated normal path $\Gamma_{n,N} : Z_n^{(N)} \rightsquigarrow X_n$ satisfying

$$S_0(\Gamma_{n,N}) + S_1(\Gamma_{n,N}) \leq Cr_N. \quad (50)$$

If $N \geq n+1$, the same path also satisfies

$$S_2(\Gamma_{n,N}) \leq Cr_N. \quad (51)$$

The orbit is unique among exact inverse orbits that remain in the C^2 closures of the shrinking tubes constructed in the proof.

There is also a number $\eta_* = \eta_*(\kappa_0) > 0$ such that no initial indices need be discarded when $r_0 \leq \eta_*$.

Proof. Defect paths and tube constants. Let $C_{\text{int}} = C_{\text{int}}(\kappa_0)$ be the constant in Lemma 6.2, put

$$d_n = C_{\text{int}}e_n, \quad a_n = \sum_{m=n}^{\infty} d_m = C_{\text{int}}r_n.$$

Lemma 6.2 gives a defect path $\Lambda_n : Q_n \rightsquigarrow A_n$ satisfying

$$0 < \kappa \leq \kappa_0, \quad W(\Lambda_n) + S_0(\Lambda_n) + S_1(\Lambda_n) \leq d_n.$$

Set

$$\bar{\kappa} = \frac{2\kappa_0 + 1}{3}, \quad \hat{\kappa} = \frac{\kappa_0 + 2}{3},$$

so that $\kappa_0 < \bar{\kappa} < \hat{\kappa} < 1$. Let C_0, C_1, C_2 be the constants in Lemma 6.4 for the ceiling $\hat{\kappa}$, and set

$$M_0 = \max\{1, C_0\}, \quad M_1 = \max\{1, C_1(1 + M_0)\}. \quad (52)$$

Define

$$\begin{aligned} C_{\text{tube}} &= C_2(1 + M_0 + M_1) + \hat{\kappa}^2 C_0, \\ C_{\text{inc}} &= C_2(1 + M_0 + M_1) + \hat{\kappa}^2 M_0. \end{aligned} \quad (53)$$

All these constants depend only on κ_0 . Define $\mathcal{D}_n$ to be the set of curves that can be joined to Q_n by a piecewise smooth path satisfying

$$\begin{aligned} 0 < \kappa &\leq \hat{\kappa}, \\ W &\leq a_n, \quad S_0 \leq M_0 a_n, \quad S_1 \leq M_1 a_n. \end{aligned}$$

Invariant tubes. Discard finitely many initial indices so that

$$C_{\text{tube}} a_{n+1} < \bar{\kappa} - \kappa_0 \quad (54)$$

for every remaining n . We prove that

$$\mathcal{B}(\mathcal{D}_{n+1}) \subseteq \mathcal{D}_n. \quad (55)$$

Let $C \in \mathcal{D}_{n+1}$, and let Γ be an admissible path from Q_{n+1} to C . By Lemma 6.3, $\mathcal{B}\Gamma$ is a smooth path. Lemma 6.4 gives

$$\begin{aligned} W(\mathcal{B}\Gamma) &\leq a_{n+1}, \\ S_0(\mathcal{B}\Gamma) &\leq C_0 a_{n+1}, \\ S_1(\mathcal{B}\Gamma) &\leq C_1(1 + M_0)a_{n+1}. \end{aligned}$$

The second-derivative estimate gives

$$S_2(\mathcal{B}\Gamma) + \hat{\kappa}^2 S_0(\mathcal{B}\Gamma) \leq C_{\text{tube}} a_{n+1}.$$

The output starts at A_n , whose curvature is at most κ_0 . Equations (54) and Lemma 6.6 keep the entire inverse path below $\bar{\kappa}$, hence below $\hat{\kappa}$. Concatenate it with Λ_n . Since $a_n = d_n + a_{n+1}$,

$$\begin{aligned} W &\leq a_n, \\ S_0 &\leq d_n + C_0 a_{n+1} \leq M_0 a_n, \\ S_1 &\leq d_n + C_1(1 + M_0)a_{n+1} \leq M_1 a_n. \end{aligned}$$

These inequalities prove (55). They also give the endpoint margin

$$\max \kappa_C \leq \kappa_0 + C_{\text{tube}} a_{n+1} \leq \bar{\kappa} \quad (56)$$

for every $C \in \mathcal{B}(\mathcal{D}_{n+1})$. Every finite terminal pullback therefore exists and lies in its corresponding tube.

Cauchy convergence. Fix n and compare terminal indices N and $N + 1$. The path Λ_N joins Q_N to $A_N = \mathcal{B}Q_{N+1}$, and $\mathcal{B}^{N-n}\Lambda_N$ joins $Z_n^{(N)}$ to $Z_n^{(N+1)}$. Before any inverse is applied,

$$W + S_0 + S_1 \leq d_N.$$

If $N \geq n + 1$, the first inverse step gives

$$\begin{aligned} W &\leq d_N, \quad S_0 \leq M_0 d_N, \quad S_1 \leq M_1 d_N, \\ S_2 &\leq C_2(1 + M_0 + M_1)d_N. \end{aligned}$$

The same bounds hold after every subsequent inverse step. At each depth, the output path starts at a finite pullback whose curvature is at most $\bar{\kappa}$. Discard finitely many more levels so that

$$C_{\text{inc}} d_N < \hat{\kappa} - \bar{\kappa}$$

for every remaining N . Lemma 6.6 then keeps every propagated path below $\hat{\kappa}$. Consequently, for every $N \geq n$, the increment path satisfies

$$S_0 + S_1 \leq C d_N, \quad (57)$$

and, when $N \geq n + 1$, it also satisfies

$$S_2 \leq C d_N. \quad (58)$$

Fix n . Begin the Cauchy argument with terminal indices $N \geq n + 1$. Equations (57)–(58) and $\sum d_N < \infty$ satisfy the hypotheses of Lemma 6.5; denote the resulting limit by X_n . Concatenating the increment paths gives (50). If $N \geq n + 1$, it also gives (51). Every finite pullback at level n is joined to Q_n by a defining tube path with $S_0 \leq M_0 a_n$ and $W \leq a_n$. Passing to the limit gives

$$d_H(X_n, Q_n) \leq M_0 a_n, \quad |\text{Per}(X_n) - 2L_n| \leq \hat{\kappa} a_n;$$

the second inequality is the absolute first-variation estimate for perimeter along a normal path. Since $a_n = C_{\text{int}} r_n$, this proves (49). Continuity of $\mathcal{T}\gamma = \gamma + \tau_\gamma$ in C^1 gives $\mathcal{T}X_n = X_{n+1}$.

Convexity and regularity. Every finite terminal pullback is strictly convex and has turning number one. For each n , let $\delta_{n,N}$ and $\sigma_{n,N}$ denote the steering angle and the inverse rear-arclength correspondence of the terminal pullback $Z_n^{(N)}$. By the Lipschitz bounds of Lemma 6.7, the families $\{\delta_{n,N}\}_N$ and $\{\sigma_{n,N}\}_N$ are uniformly bounded and equicontinuous, so a diagonal Arzelà–Ascoli extraction gives one subsequence of terminal indices along which both converge uniformly for every n ; the limits satisfy the same steering ODE and right-inverse identity. Marked geometric C^2 convergence, strengthened in this way to C^0 convergence of the steering data, therefore implies

$$\kappa_{X_n} \geq 0.$$

The unit tangents converge uniformly as maps $S^1 \rightarrow S^1$. For all large indices, each approximating tangent is joined to the limiting tangent by the shorter arc of S^1 ; hence the limiting tangent still has degree one. Its tangent angle is nondecreasing and increases by 2π in one circuit.

A regular closed plane curve with nonnegative curvature and turning number one bounds a convex body: in each direction, its scalar projection has one interval of increase and one of decrease, and therefore attains its maximum on a supporting line. Thus X_n is convex, and its curvature is at most $\bar{\kappa} < 1$.

We establish smoothness before upgrading convexity to strict convexity. Along every finite rear–front pair the steering angle satisfies $0 < \delta \leq \arctan \bar{\kappa}$. The curves and their unit tangents converge, so the limiting pair has a steering angle in the closed strip

$$0 \leq \delta \leq \arctan \bar{\kappa} < \frac{\pi}{2}.$$

Lemma 6.7 shows that X_n is the unique regular-branch selected rear of X_{n+1} . Thus $X_n = \mathcal{B}X_{n+1}$ also holds at the limit. Iterating the regularity gain in $X_n = \mathcal{B}^r X_{n+r}$ makes every X_n smooth. Apply Lemma 2.2 to the smooth consecutive convex curves X_n and $X_{n+1} = \mathcal{T}X_n$. Every X_n is strictly convex.

Uniqueness. Let (Y_n) and $(\tilde{Y}_n)$ be two exact inverse orbits with $Y_N, \tilde{Y}_N \in \overline{\mathcal{D}_N}^{C^2}$ for every large N . Approximate the two curves at level N by elements of $\mathcal{D}_N$, concatenate their admissible paths through Q_N , and propagate the resulting comparison path backward. The same estimates give C^2 -distance at most $C a_N$ at every fixed earlier level. First pass to the approximating limit, using Lemma 6.7 at each of the finitely many inverse steps. Then let $N \rightarrow \infty$. Since $a_N \rightarrow 0$, the two exact orbits coincide.

Uniform smallness. Only two smallness conditions require the deletion of initial indices: (54) and the propagated-increment inequality. Since $d_n \leq a_0 = C_{\text{int}} r_0$, the choice

$$\eta_* = \frac{1}{2C_{\text{int}}} \min \left\{ \frac{\bar{\kappa} - \kappa_0}{C_{\text{tube}}}, \frac{\hat{\kappa} - \bar{\kappa}}{C_{\text{inc}}} \right\} \quad (59)$$

makes both inequalities valid from level zero whenever $r_0 \leq \eta_*$. $\square$

The shadowing theorem is independent of the hairpin construction: it applies to any low-curvature inverse pseudo-orbit with summable intrinsic-curvature defects.

7 Proof of the main theorem

Lemma 7.1 (Large-separation threshold). *For the fixed translating hairpin, there is $H_* < \infty$ such that every initial separation $H_0 \geq H_*$ satisfies the properties below.*

- i. *The map P is strictly increasing on the entire cap sequence, $P(H) \leq H - \Delta/2$, and the recursion $P(H_{n+1}) = H_n$ has a unique solution with*

$$H_n \geq H_0 + \frac{\Delta}{2}n.$$

- ii. *Proposition 4.3, Lemma 4.4, and Theorem 5.2 apply at every level with one common curvature ceiling $\kappa_0 < 1$.*

- iii. *If e_n and r_n are the defect and tail quantities in Theorem 6.8, then*

$$r_0 \leq C(1 + H_0)^2 e^{-\beta' H_0} \longrightarrow 0 \quad (H_0 \rightarrow \infty). \quad (60)$$

In particular, after increasing H_ , one has $r_0 \leq \eta_*(\kappa_0)$, so the shadowing theorem applies from level zero.*

- iv. *Let C_{sh} be the constant in (49) and C_W the constant in Lemma 4.4. Increase H_* , if necessary, so that*

$$\frac{2H_0 - C_{\text{sh}}r_0}{\pi} > C_W + 2C_{\text{sh}}r_0. \quad (61)$$

Proof. Take H_* larger than the finitely many geometric thresholds in Proposition 4.3, Lemma 4.4, and Theorem 5.2, and so large that

$$P'(H) \geq \frac{1}{2}, \quad |P(H) - (H - \Delta)| \leq \frac{\Delta}{2}.$$

These inequalities make P invertible on $[H_*, \infty)$ and imply the stated growth of H_n . After decreasing the exponent if necessary, the matching theorem gives

$$e_n \leq C(1 + H_n)^2 e^{-\beta H_{n+1}}.$$

The bound $H_n \geq H_0 + n\Delta/2$ reduces the sum to a polynomially weighted geometric series and proves (60). Since its right side tends to zero, the third assertion follows from the uniformity clause in Theorem 6.8. The left side of (61) tends to infinity, while its right side tends to C_W ; one final increase of H_* proves the fourth assertion. $\square$

Choose $H_0 \geq H_*$ as in Lemma 7.1. *Construction of the pseudo-orbit.* Define recursively

$$P(H_{n+1}) = H_n. \quad (62)$$

Then

$$H_{n+1} = H_n + \Delta + O(e^{-\beta H_n}), \quad H_n \geq H_0 + \frac{\Delta}{2}n.$$

Choose compatible representatives by Lemma 6.1. Fix the centered orientation and marked phase of $Q_0 = F_{H_0}$. Inductively, rotate the entire exact pair $R_{H_{n+1}} \mapsto F_{H_{n+1}}$ so that the marked tangent of $R_{H_{n+1}}$, at the phase used in Theorem 5.2, agrees with the marked tangent of Q_n . Put

$$Q_{n+1} = F_{H_{n+1}}$$

with this rotation, and set

$$A_n = \mathcal{B}Q_{n+1} = R_{H_{n+1}}.$$

Central symmetry keeps every representative centered at the origin, and rotation does not alter any curvature or matching estimate. Hence Q_n and A_n have the same marked tangent phase and the same half-perimeter H_n . The matching theorem gives

$$\|\kappa_{Q_n} - \kappa_{A_n}\|_{L^1(0, H_n)} \leq C e^{-\beta H_{n+1}}.$$

Therefore the quantities e_n, r_n of Theorem 6.8 satisfy (60). The common curvature ceiling follows from Proposition 4.3, and the choice of H_0 ensures $r_0 \leq \eta_*(\kappa_0)$. The shadowing theorem therefore applies from level zero and gives a smooth exact orbit

$$\mathcal{T}X_n = X_{n+1}.$$

Excluding a circle. By Lemma 4.4, the transverse width of $Q_0 = F_{H_0}$ is bounded uniformly for all sufficiently large H_0 , whereas $\text{Per}(Q_0) = 2H_0$. The stated estimate (49) and Lemma 7.1 show that

$$d_H(X_0, Q_0) + |\text{Per}(X_0) - 2H_0| \leq Cr_0 \longrightarrow 0$$

as the initial separation $H_0 \rightarrow \infty$. A Hausdorff perturbation of size d changes every directional width by at most $2d$. Therefore the transverse width of X_0 is at most $C_W + 2C_{\text{sh}}r_0$, whereas a circle with this perimeter would have width at least $(2H_0 - C_{\text{sh}}r_0)/\pi$. This contradicts (61) and completes the proof of Theorem 1.1.

A Numerically generated geometry

The figures are numerical illustrations and play no role in the proof. Their coordinates were generated from the six-mode cosine profile with $\Lambda = 25.0400224235038$. The corresponding translation is $V \approx 0.01945183447$; the cap-pair plot uses $H = 160$.

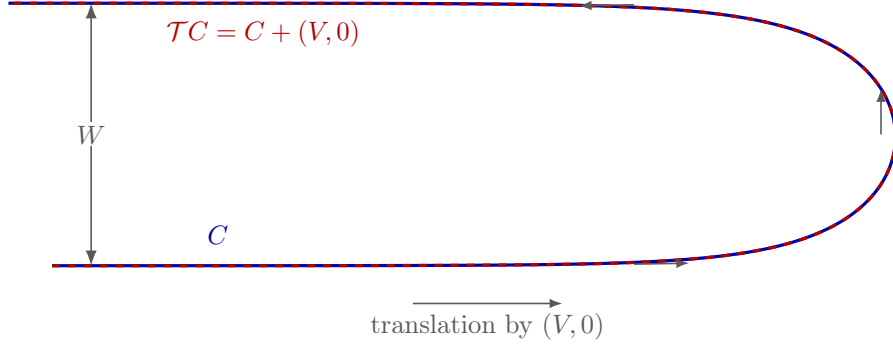


Figure 1: The sampled hairpin profile and its horizontal translate by $V \approx 0.01945183447$. The solid curve is C ; the dashed curve is $C + (V, 0)$. The arrows show the positive tangent direction, and W marks the strip width.

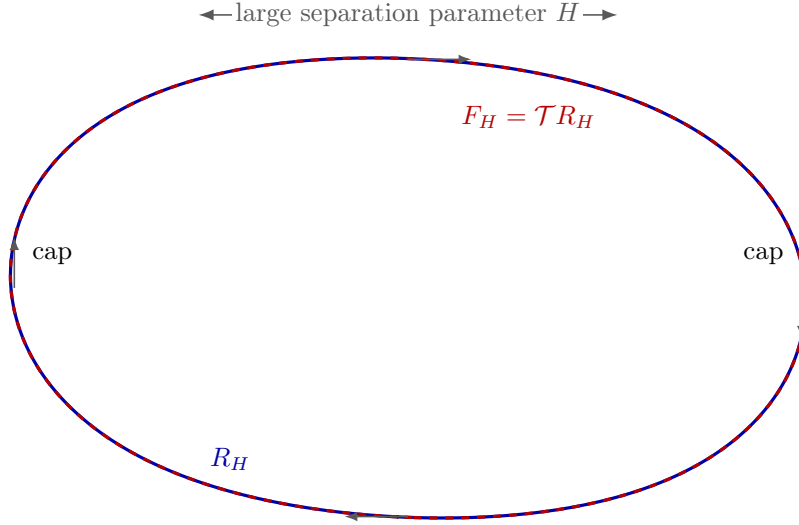


Figure 2: A numerical reconstruction of the periodized rear-front pair at $H = 160$. The solid curve is R_H ; the dashed curve is $F_H = \mathcal{T}R_H$, up to numerical quadrature error. The two labeled ends are the cap regions.

References

- [1] G. Bor, M. Levi, R. Perline, and S. Tabachnikov, Tire tracks and integrable curve evolution, *International Mathematics Research Notices* **2020**, no. 9, 2698–2768, [doi:10.1093/imrn/rny087](https://doi.org/10.1093/imrn/rny087).
- [2] R. L. Foote, Geometry of the Prytz planimeter, *Reports on Mathematical Physics* **42** (1998), no. 1–2, 249–271, [doi:10.1016/S0034-4877\(98\)80013-X](https://doi.org/10.1016/S0034-4877(98)80013-X).
- [3] R. L. Foote, M. Levi, and S. Tabachnikov, Tractrices, bicycle tire tracks, hatchet planimeters, and a 100-year-old conjecture, *American Mathematical Monthly* **120** (2013), no. 3, 199–216, [doi:10.4169/amer.math.monthly.120.03.199](https://doi.org/10.4169/amer.math.monthly.120.03.199).
- [4] M. Levi and S. Tabachnikov, On bicycle tire tracks geometry, hatchet planimeter, Menzin’s conjecture, and oscillation of unicycle tracks, *Experimental Mathematics* **18** (2009), no. 2, 173–186, [doi:10.1080/10586458.2009.10128894](https://doi.org/10.1080/10586458.2009.10128894).
- [5] S. Tabachnikov, Tire track geometry: variations on a theme, *Israel Journal of Mathematics* **151** (2006), 1–28, [doi:10.1007/BF02777353](https://doi.org/10.1007/BF02777353).
- [6] S. Tabachnikov, Iterating skew evolutes and skew involutes: a linear analog of bicycle kinematics, *Moscow Mathematical Journal* **23** (2023), no. 4, 625–640, [doi:10.17323/1609-4514-2023-23-4-625-640](https://doi.org/10.17323/1609-4514-2023-23-4-625-640).